

MAINCRAFTS TECHNOLOGY

Cybersecurity & Ethical Hacking Internship Program

THREAT INTELLIGENCE REPORT

Task-1: Awareness & Research Project

Modern Cyber Threats (2024-2025) and Defense Strategies

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1. Introduction to Cybersecurity

1.1 What is Cybersecurity?

At its core, cybersecurity is about keeping computers, networks, phones, apps, and data safe from people who want to steal, damage, or misuse them. It is not just antivirus software or a firewall — it is a mix of technology, good processes, and, honestly, a lot of human awareness. Attackers range from bored teenagers experimenting in their bedrooms to full-blown organized crime groups and even nation-states, so “defense” has to cover a very wide spectrum of skill and intent. Most textbooks explain the goal of cybersecurity using the CIA triad: Confidentiality (only the right people can see the data), Integrity (the data hasn't been tampered with), and Availability (the systems are actually up and usable when you need them). It sounds simple written down, but getting all three right at once, all the time, is genuinely hard.

1.2 Why is it Important for Individuals and Businesses?

For an individual, poor cybersecurity might mean someone drains your bank account, takes over your social media, or gets hold of private photos and messages. It's personal, and the damage isn't only financial — there's a real emotional toll too. For a business, the consequences scale up fast. A breach can shut down operations for days, leak thousands (or millions) of customer records, invite regulatory fines, and quietly chip away at a brand's reputation long after the headlines fade. And because almost every business function today runs through some digital system — payroll, customer support, manufacturing lines, you name it — there are simply more doors for an attacker to try. That's why cybersecurity has stopped being “an IT problem” and become a board-level business risk.

1.3 Current Relevance: Cybercrime, Digital Dependency, and AI-Driven Threats

Three things stand out about the threat landscape right now, in 2024–2025. First, cybercrime just keeps growing — the projected global cost is now measured in trillions of dollars a year, which is hard to even picture. Second, we've become deeply dependent on digital systems that weren't always built with security as the top priority; hospitals, banks, and power grids increasingly run on cloud platforms and connected devices that can be someone's weekend side-project turned mission-critical infrastructure. Third — and this is the part that feels genuinely new — AI has started working for the attackers too. Convincing phishing emails, cloned voices, and even fake video calls can now be produced in minutes instead of hours, and that's changing the game faster than most defenses can keep up.

82.6% of phishing emails now contain some form of AI-generated content	1,265% surge in phishing attacks linked to generative AI since 2023	\$2.4B Change Healthcare ransomware breach cost UnitedHealth Group
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Sources: Hoxhunt/Brightside AI Phishing Trends analysis (2025–2026); PMC health-policy study on the Change Healthcare ransomware attack (2024).

2-5. Major Modern Cyber Threats: Impact, Case Studies & Defenses

Below are five threats that came up again and again while researching this report. For each one, I've looked at who gets hurt (individuals vs. organizations), pulled in a real incident from 2024-2025 to show what it actually looks like in practice, and listed a few defenses that would have realistically helped.

2.1 Threat 1: AI-Powered Phishing Attacks

This is probably the threat that surprised me the most while researching. AI-powered phishing doesn't look like the old “prince needs your bank details” emails anymore. Attackers now use large language models to write flawless, personalized messages, and they've started using voice cloning and even deepfake video to impersonate real people on calls. The old advice — “look for typos and weird grammar” — basically doesn't work anymore.

Impact on Individuals	Credential theft, drained bank accounts, and fraudulent authorizations obtained through convincing fake calls or emails impersonating trusted contacts, banks, or employers.
Impact on Organizations	Large fraudulent wire transfers, business email compromise, compromised employee credentials leading to network intrusion, and reputational damage when customers are targeted using the company's name.

CASE STUDY: The Arup Deepfake Video Conference Fraud (February 2024)

Company: Arup (UK-based engineering firm) | Loss: US \$25 million

A finance employee at Arup, the multinational engineering firm behind projects such as the Sydney Opera House renovation, joined what appeared to be a routine video call with the company's CFO and several senior colleagues. Every face and voice on the call was an AI-generated deepfake built from publicly available footage of the real executives. Believing the instructions were legitimate, the employee authorized transfers totaling \$25 million to attacker-controlled accounts. The incident is one of the largest publicly confirmed deepfake-enabled fraud cases and illustrates how AI has made even face-to-face verification unreliable.

Preventive Measures

- Enforce out-of-band verification for any financial request (e.g., a callback to a known, pre-verified phone number) regardless of how convincing a video or voice call appears.
- Deploy AI-aware email security and deepfake-detection tools alongside Multi-Factor Authentication (MFA) on all financial systems.
- Run regular Security Awareness Training that specifically covers AI-generated phishing, voice cloning, and deepfake video scenarios — not just traditional email red flags.

2.2 Threat 2: Ransomware-as-a-Service (RaaS)

Ransomware used to require some real technical skill to pull off. Not anymore. RaaS groups now build the malware and the infrastructure, then rent it out to “affiliates” for a cut of whatever ransom gets paid — basically a criminal franchise model. This is a big reason ransomware attacks on hospitals, schools, and city

governments have kept climbing; you no longer need to be a skilled hacker to launch one, just willing to pay for access.

Impact on Individuals	Exposure of highly sensitive personal and medical data, delayed access to healthcare or financial services, and downstream identity-theft risk from leaked records.
Impact on Organizations	Multi-day to multi-month operational shutdowns, ransom payments in the millions of dollars, regulatory fines, class-action lawsuits, and long-term reputational damage.

CASE STUDY: Change Healthcare / UnitedHealth Group Ransomware Attack (February 2024)

Threat actor: BlackCat/ALPHV (RaaS) | Cost: over US \$2.4 billion | Records exposed: ~190 million individuals

On February 21, 2024, the BlackCat/ALPHV ransomware-as-a-service group gained access to Change Healthcare's network through a remote access portal that lacked multi-factor authentication, using compromised credentials. The attackers exfiltrated vast amounts of protected health information before deploying ransomware, forcing UnitedHealth Group to disconnect Change Healthcare's systems entirely to contain the damage. The resulting outage disrupted prescription processing and insurance claims across the U.S. healthcare system for months. UnitedHealth Group paid a reported \$22 million ransom, and total incident costs have since risen above \$2.4 billion, making it the largest healthcare data breach in U.S. history and a textbook example of how a single missing security control — MFA on a remote access point — can cascade into a national-scale crisis.

Preventive Measures

- Mandate Multi-Factor Authentication (MFA) on every remote access point, VPN, and privileged account without exception.
- Maintain immutable, regularly tested offline backups to enable recovery without paying a ransom.
- Adopt a Zero Trust Security Model and network segmentation to limit how far an intrusion can spread once inside the network.

2.3 Threat 3: Cloud Security Misconfigurations

What struck me researching this one is that most cloud breaches aren't really “hacks” at all — they're mistakes. A storage bucket left public, a permission set too broadly, encryption that was never turned on. AWS, Azure, and Google Cloud all provide the tools to lock things down properly, but the responsibility for actually using them sits with whoever configures the account. Gartner has even predicted that by 2025, nearly all cloud security failures will trace back to the customer's setup, not the cloud provider itself.

Impact on Individuals	Personal records — including résumés, health data, and financial details — left openly accessible on the internet without any hacking required, enabling identity theft and targeted scams.
Impact on Organizations	Mass data exposure, regulatory fines under laws such as GDPR or HIPAA, loss of customer trust, and costly incident response and legal proceedings.

CASE STUDY: TalentHook Azure Blob Exposure (2024)

Company: TalentHook (resume-building platform) | Records exposed: 26 million+

TalentHook, a resume-building platform, left an Azure Blob storage container open to public access without authentication. Over 26 million résumés — containing names, email addresses, phone numbers, work history, and education details — were left freely accessible to anyone who found the container. No sophisticated hacking technique was required; the data was exposed purely because of a basic access-control misconfiguration, echoing the pattern seen in the earlier and better-known 2019 Capital One breach, where a misconfigured web application firewall on AWS enabled attackers to access the data of over 100 million customers.

Preventive Measures

- Enforce the principle of least privilege on all IAM roles and regularly audit permissions using tools such as AWS IAM Access Analyzer or Azure Policy.
- Enable public-access blocks by default on all storage buckets/containers and require explicit justification to make any asset public.
- Use Cloud Security Posture Management (CSPM) tools to continuously scan for and auto-remediate misconfigurations across multi-cloud environments.

2.4 Threat 4: IoT Vulnerabilities

Smart cameras, routers, DVRs — most of us have a handful of these at home, and most of us have never changed the default password on any of them. That's exactly the gap attackers exploit. IoT devices are often built cheaply, ship with weak or hardcoded credentials, and rarely get security updates. Once compromised, a device usually doesn't get “hacked” in any dramatic sense — it just quietly gets recruited into a botnet and used to attack someone else.

Impact on Individuals	Compromised home routers or cameras can be used to spy on residents, intercept network traffic, or silently participate in criminal botnets without the owner's knowledge.
Impact on Organizations	Botnet-driven DDoS attacks can take down corporate websites and services for hours or days, while compromised IoT devices on a corporate network provide attackers a stealthy entry point.

CASE STUDY: Aisuru / Kimwolf IoT Botnet DDoS Campaign (2024–2026)

Devices compromised: 3 million+ (routers, IP cameras, DVRs, smart-TV boxes) | Peak DDoS: 20+ Tbps

The Aisuru botnet emerged in late 2024 by exploiting weak credentials and unpatched firmware in home routers, IP cameras, and DVRs, rapidly growing into one of the most powerful DDoS platforms ever observed, with attack capacity exceeding 20 terabits per second by 2025–2026. A related variant, Kimwolf, later added the ability to infect devices hidden behind home network address translation (NAT), expanding the botnet to more than 3 million compromised devices worldwide. In March 2026, the U.S. Department of Justice seized the botnet's command-and-control infrastructure, but security researchers noted that the underlying vulnerable devices remained unpatched and were quickly targeted by copycat botnets — highlighting that takedowns alone cannot fix the root cause of insecure IoT hardware.

Preventive Measures

- Change all default usernames and passwords on IoT devices immediately after purchase and use unique, strong credentials.

- Keep device firmware updated automatically where possible, and retire devices that no longer receive security patches.
- Segment IoT devices onto a separate network (VLAN) isolated from critical business or personal systems, and disable unused remote-access features.

2.5 Threat 5: Zero-Day Exploits

A zero-day is basically the worst-case scenario in security: a flaw nobody knows about yet, including the company that made the software. There's no patch, because there's nothing to patch — the vendor doesn't even know the hole exists. Attackers who find these first can quietly exploit them for months before anyone notices, which is exactly what makes zero-days so dangerous when they hit software that thousands of organizations rely on.

Impact on Individuals	Personal and financial data held by any organization using the vulnerable software can be silently stolen long before the flaw is even discovered, let alone patched.
Impact on Organizations	Mass, simultaneous compromise across thousands of organizations that use the same software; costly emergency patching; regulatory reporting obligations; and supply-chain-wide reputational damage.

CASE STUDY: MOVEit Transfer Zero-Day Exploited by Cl0p (May 2023 – ongoing disclosures through 2024)

CVE-2023-34362 (CVSS 9.8) | Organizations affected: 2,500+ confirmed, ~8,000 estimated globally

The Cl0p ransomware group discovered and exploited a previously unknown SQL injection vulnerability in Progress Software's MOVEit Transfer file-sharing platform, deploying a custom web shell named LEMURLOOT to steal data directly from victims' databases. Because MOVEit was widely used by banks, universities, airlines, payroll providers, and government agencies to transfer sensitive files, a single zero-day flaw cascaded into one of the largest supply-chain data breaches on record, with confirmed impact on more than 2,500 organizations and tens of millions of individuals. Notably, Cl0p had been quietly testing the vulnerability since mid-2021, well before the mass-exploitation campaign began — illustrating how zero-days can be stockpiled by attackers long before their impact becomes visible.

Preventive Measures

- Implement rigorous, rapid Patch Management processes so emergency patches can be deployed within hours of release, not weeks.
- Deploy Intrusion Detection/Prevention Systems (IDS/IPS) and Web Application Firewalls (WAF) to catch anomalous exploitation attempts even before a patch exists.
- Minimize the internet-facing attack surface of critical software (e.g., file-transfer and remote-access tools) and monitor vendor security advisories continuously.

Threat Summary at a Glance

Threat	Primary Risk	Real-World Case	Key Defense
AI-Powered Phishing	Deepfake-driven fraud	Arup (\$25M, 2024)	Out-of-band verification
Ransomware-as-a-Service	Operational shutdown	Change Healthcare (2024)	MFA + offline backups
Cloud Misconfiguration	Mass data exposure	TalentHook (2024)	Least-privilege IAM + CSPM
IoT Vulnerabilities	Botnet recruitment / DDoS	Aisuru/Kimwolf botnet	Change defaults + segmentation
Zero-Day Exploits	Undetected mass breach	MOVEit/ClOp (2023–24)	Rapid patching + IDS/IPS

6. Conclusion & Future Scope

6.1 Why Proactive Cybersecurity Matters

Looking back at all five threats — AI phishing, RaaS, cloud misconfigurations, IoT weaknesses, and zero-days — one pattern kept showing up: it's rarely some brilliant, unstoppable hack. It's a missing MFA setting, a default password nobody changed, a permission that was too broad. Change Healthcare's breach traces back to one remote-access portal without MFA. TalentHook's leak traces back to a storage container that was never locked down. That's actually a hopeful takeaway, in a way — it means most of these breaches were preventable with fairly ordinary security hygiene: MFA everywhere, least-privilege access, patching on time, and training people to double-check before they click or approve something. Waiting until after a breach to react just isn't good enough anymore given how fast these attacks move.

6.2 The Need for Continuous Learning

If there's one thing I'd want to remember from doing this task, it's that cybersecurity isn't something you learn once and finish. AI is being used by both sides right now — attackers are using it to write better phishing emails, and defenders are using it to catch things faster — and that back-and-forth isn't going to settle down anytime soon. New vulnerabilities show up constantly, and a defense that worked great last year might already have a gap in it today. Going forward, I think the most useful habit I can build isn't memorizing this year's threats, but staying in the habit of checking sources like CISA and OWASP regularly, practicing hands-on, and just staying curious about how attackers are adapting — because they definitely are.

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